Instituto Superior Técnico

BSc Aerospace Engineering

Multivariable, Nonlinear and Optimal Control

Final Report: Linear State Feedback Control of an Inverted Pendulum

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I. Introdução

A. Context and Objectives

This report details the work developed within the scope of the Multivariable, Non-linear, and Optimal Control course laboratory project. The primary objective is to design, simulate, and implement a robust control strategy to stabilize a highly non-linear and inherently unstable system using a pre-existing experimental setup of a rotary inverted pendulum.

The adopted methodology relies on linearizing the plant's mathematical model around its unstable equilibrium point (the upward vertical position) to derive a Linear Time-Invariant (LTI) state-space representation. Based on this model, a Linear Quadratic Gaussian (LQG) control architecture is developed. This comprises a Linear Quadratic Regulator (LQR) to compute the optimal state feedback law, and a Linear Quadratic Estimator (LQE, or Kalman Filter) to estimate the system's full internal state from noisy measurements. The project covers the theoretical analysis of the system, validation through MATLAB/Simulink simulations, and practical real-time implementation on the laboratory's physical hardware, specifically addressing real-world non-linearities such as motor deadzones and steady-state errors.

B. System Description

The physical plant utilized in this work is a rotary inverted pendulum apparatus (a Quanser experimental setup). The mechanical structure consists of a motorized rotary arm that actively maneuvers an unactuated, free-swinging pendulum rod attached to its extremity.

The system operates as a Single-Input Multiple-Output (SIMO) process. The single control input u is the continuous voltage applied to the DC motor that drives the horizontal rotary arm. To monitor the system's behavior, two physical sensors provide the measured output vector y:

- A potentiometer measuring the angular position of the rotary arm α.
- An optical encoder measuring the precise angular position of the pendulum relative to the arm β.

Because the system's complete state vector includes both the angular positions α, β and their respective angular velocities $\dot{\alpha}, \dot{\beta}$, but only the positions are directly measured, the system requires a state observer. Furthermore, the provided physical setup interfaces with the Simulink control environment via A/D and D/A data acquisition boards, allowing the designed controllers to be executed and tested on the real system in real-time.

II. Theoretical Analysis and Control Design

A. Space-State Formulation

A dinâmica não-linear do distema implica que este necessite de ser linearizado em torno de um ponto de equilíbrio, que dado o objetivo deste relatório, é o ponto de equilíbrio instável onde o pêndulo se encontra na posição vertical para cima. Este equilíbrio corresponde a todas as variáveis de estado iguais a zero. A derivação do modelo não-linear e a subsequente linearização não são objeto deste relatório, podendo ser encontrados no Capítulo 2 da tese de Samuel Balula **(fazer referência)**. O modelo linearizado do sistema que representa a dinâmica razoavelmente bem é dado por:

$$\begin{cases} \dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \\ \mathbf{y} = C\mathbf{x} + D\mathbf{u} \end{cases}$$

onde $\mathbf{x} = [\alpha\ \dot{\alpha}\ \beta\ \dot{\beta}\ i]^T$ é o vetor de estados de dimensão 5 e i é a corrente do motor. O vetor de entrada unidimensional $\mathbf{u} = u$ corresponde à diferença de potencial contínua, em *Volts*, aplicada nos terminais do motor DC através de um power amplifier, controlando diretamente α. O vetor de saída $\mathbf{y} = [\alpha\ \beta]^T$ é composto pelas medições dos sensores, correspondendo às posições angulares do braço e do pêndulo, respetivamente.

As matrizes do modelo A, B, C e D

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.0174 & 20.7861 & -0.0023 & 57.5344 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -0.0174 & 63.8319 & -0.0071 & 57.4388 \\ 0 & -232.0252 & 0 & 0 & -755.425 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 333.3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

B. State-Space Analysis

In this section, we analyze the previously formulated state-space model in terms of system poles, controllability, observability, and frequency response.

1) Q1 - System Poles: The system poles are the eigenvalues of the dynamics matrix A. Using MATLAB's `eig()` function, the following poles were obtained:

Tabela I: Características dos Modos do Sistema

Polos	ω (rad/s)	τ (s)
0	0	∞
-737.3	7.37×10^2	1.36×10^{-3}
-19.36	1.94×10^1	5.16×10^{-2}
-5.761	5.76	1.74×10^{-1}
6.997	7.00	-1.43×10^{-1}

Analysis of the poles shows that the system has three stable modes: two correspond to the mechanical damping of the pendulum, $\lambda_3 = -19.3673$ and $\lambda_4 = -5.7612$, and a very fast mode associated with the motor current, $\lambda_2 = -737.3184$. Additionally, the state α generates a pole at the origin, $\lambda_1 = 0$, due to being an integrator. This implies that after a perturbation, α will accumulate error over time without diverging or converging, reflecting the fact that the arm rotates without friction and does not naturally settle to an equilibrium.

The single unstable mode, $\lambda_5 = 6.9974$, is related to the vertical pendulum position, which is the primary state we aim to control.

2) Q2 - Controllability: A system is controllable if it is possible to transition from any state x_1 to another arbitrary state x_2 by applying an input $u(t)$ over a finite time interval. Knowing whether the system is controllable is crucial for understanding the limitations of the controller.

The controllability matrix is defined as:

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}, \tag{1}$$

where n is the number of states. Using MATLAB's `ctrb(A,B)` function, we obtained:

$$\mathcal{C} = \begin{bmatrix} 0 & 0 & 1.918 \times 10^4 & -1.449 \times 10^7 & 1.069 \times 10^{10} \\ 0 & 1.918 \times 10^4 & -1.449 \times 10^7 & 1.069 \times 10^{10} & -7.881 \times 10^{12} \\ 0 & 0 & 1.915 \times 10^4 & -1.446 \times 10^7 & 1.067 \times 10^{10} \\ 0 & 1.915 \times 10^4 & -1.446 \times 10^7 & 1.067 \times 10^{10} & -7.869 \times 10^{12} \\ 3.333 \times 10^2 & -2.518 \times 10^5 & 1.858 \times 10^8 & -1.370 \times 10^{11} & 1.010 \times 10^{14} \end{bmatrix}$$

The system is controllable if the rank of the controllability matrix equals the number of states. Using MATLAB's `rank()` function, we obtained $\text{rank}(\mathcal{C}) = 5$, which equals the number of states. Therefore, the system is controllable.

3) Q3 - Observability: A system is observable if, for any initial state $x(0)$, knowledge of the input $u(t)$ and output $y(t)$ over a finite time interval uniquely determines $x(0)$. Although the system is controllable, implementing a state-feedback controller requires knowledge of the full state[1]. Since the state is not directly measurable, it must be reconstructed from the outputs, which is only possible if the system is observable.

The observability matrix is defined as:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \tag{2}$$

Using MATLAB's `obsv()` function, the observability matrix was calculated as:

$$\mathcal{O} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -1.740 \times 10^{-2} & 2.079 \times 10^1 & -2.316 \times 10^{-3} & 5.753 \times 10^1 \\ 0 & -1.738 \times 10^{-2} & 6.383 \times 10^1 & -7.114 \times 10^{-3} & 5.744 \times 10^1 \\ 0 & -1.335 \times 10^4 & -5.096 \times 10^{-1} & 2.079 \times 10^1 & -4.346 \times 10^4 \\ 0 & -1.333 \times 10^4 & -8.152 \times 10^{-1} & 6.383 \times 10^1 & -4.339 \times 10^4 \\ 0 & 1.008 \times 10^7 & -2.762 \times 10^5 & 3.027 \times 10^1 & 3.207 \times 10^7 \\ 0 & 1.007 \times 10^7 & -2.729 \times 10^5 & 2.960 \times 10^1 & 3.202 \times 10^7 \end{bmatrix} \tag{3}$$

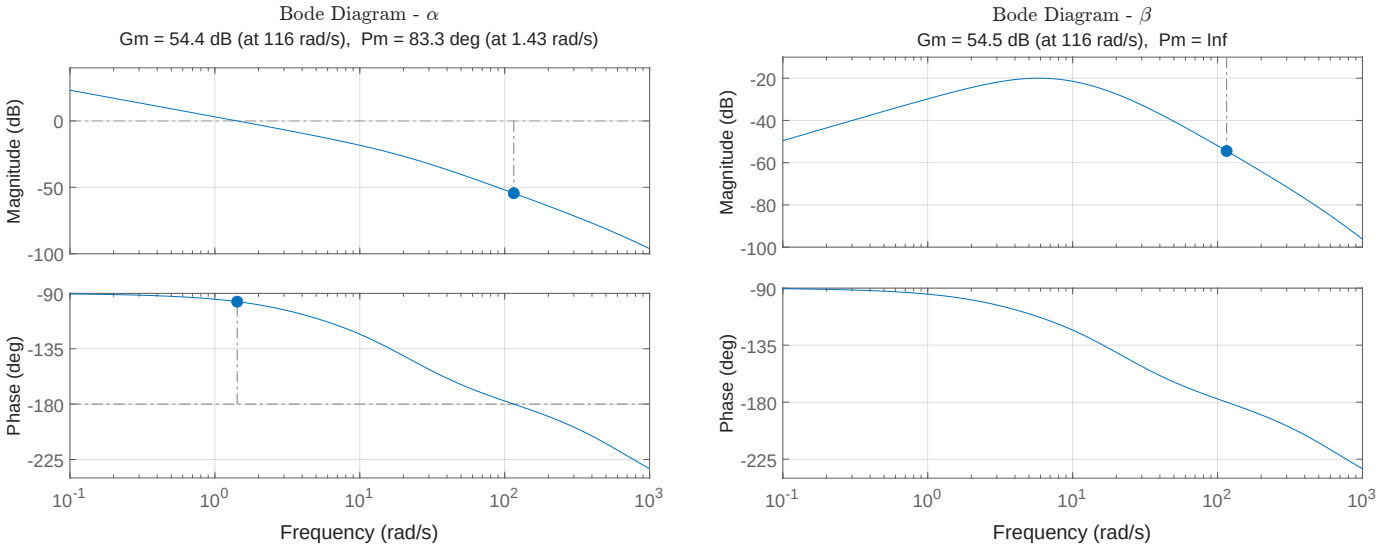
Applying the `rank()` function shows that $\text{rank}(\mathcal{O}) = 5$, which equals the number of states. Therefore, the system is observable.

4) Q4 - Bode Plots: The evaluate the frequency response of the system, the transfer functions of α and β relative to the input u were determined:

$$\frac{\alpha}{u}(s) = \frac{1.92 \times 10^4\, s^2 + 92.07\, s - 8.26 \times 10^5}{s^5 + 755.45\, s^4 + 1.33 \times 10^4\, s^3 - 4.82 \times 10^4\, s^2 - 5.76 \times 10^5\, s} \tag{4}$$

$$\frac{\beta}{u}(s) = \frac{1.92 \times 10^4\, s^2 + 92.07\, s - 8.26 \times 10^5}{s^5 + 755.45\, s^4 + 1.33 \times 10^4\, s^3 - 4.82 \times 10^4\, s^2 - 5.76 \times 10^5\, s} \tag{5}$$

[1]State-feedback control requires knowledge of the full state since the control law $u(t) = -Kx(t)$ explicitly depends on the state vector. In this project, an estimator was designed to reconstruct the state, resulting in the control law $u(t) = -K\hat{x}(t)$. State estimation is only possible if the system is observable, making observability a prerequisite for implementing state-feedback control when the state is not directly measurable.



(a) Bode plot for α (b) Bode plot for β

Figura 1: Open-loop Bode plots of the system evaluating both outputs.

C. LQR Controller Design

Q5

D. State Estimator Design - LQE

Q6, Q7

E. Baseline LQG Simulation

Q8

III. Laboratory Work and Experimental Results

A. Initial Hardware Testing

1) LQE Validation: The first stage of the experimental procedure involved the validation of the LQE. To facilitate this, the LQG controller structure was decoupled into its two primary components: the LQR regulator and the LQE observer. This separation was implemented in the Simulink model METER-MODELO by using independent blocks, as shown in the system diagram in Figure [X]. This configuration allowed for the isolation of the estimator, enabling the monitoring of the observer's performance.

The primary objective of this test was first, to ensure that the estimated states $\hat{\alpha}$ and $\hat{\beta}$ accurately tracked the raw angular measurements; and second, to verify that the remaining three unmeasured states behaved smoothly and remained strictly within expected physical boundaries, without exhibiting erratic fluctuations or excessive noise amplification. This objective was achieved by tuning the estimator's noise covariance matrices, Q_e and R_e.

A fundamental observation regarding the experimental setup was the high precision of the discrete sensors. The Quanser platform utilizes a potentiometer for the rotary arm and a high-resolution optical encoder for the pendulum angle. Due to their discrete nature, the continuous measurement noise was found to be practically negligible. However, to ensure numerical stability in the Kalman Filter algorithm and to reflect a high degree of confidence in the hardware readings, the measurement noise covariance matrix was set to:

$$R_e = I_2 \cdot (\text{deg2rad}(0.1))^2 \tag{6}$$

With R_e fixed to admit a standard deviation of only $0.1°$, the Q_e matrix was tuned empirically through hardware trials. In the Kalman Filter framework, it is the relative ratio between these covariance matrices (Q_e/R_e) that dictates the observer's gain and dynamic behavior. A high ratio yields a highly reactive filter, whereas a low ratio produces a slower, heavily filtered response.

To find the optimal balance, ensuring accurate tracking while avoiding overfitting, three distinct tests were performed by scaling the process noise identity matrix keeping R_e constant: $Q_e = 0.1 \cdot I_5$, $Q_e = I_5$, and $Q_e = 10 \cdot I_5$.

As illustrated in Figure [X], the system's real-time plots were analyzed for each configuration. The results demonstrated that while lower values of Q_e introduced a noticeable lag, the configuration with [INSERIR QUAL FOI O MELHOR] provided the best overall performance. It allowed the estimates $\hat{\alpha}$ and $\hat{\beta}$ to converge rapidly to the physical measurements with high fidelity. Crucially, the plots also confirmed that the three unmeasured states exhibited coherent dynamic trends, remaining well within the expected operational limits during the pendulum's motion. This thoroughly validated LQE setup was then adopted for all subsequent closed-loop experiments.

2) LQG Testing: Following the validation of the LQE, the complete LQG controller was implemented on the hardware. The weighting matrices for the LQR regulator, Q_r and R_r, were selected based on the optimal values determined during the theoretical simulation phase. However, for this test, we decided to maintain the default values of .....

The explicit objective of this hardware test was to demonstrate a fundamental limitation of the standard LQG architecture: its inability to stabilize the rotary arm (α) at a complete standstill at the origin.

To evaluate this, a physical disturbance was introduced to the system during operation. As illustrated in the experimental plots in Figure 2, while the controller successfully kept the pendulum upright, the rotary arm failed to return to the zero position. Instead, α exhibited a persistent steady-state error, drifting and coming to a stop away from the reference point.

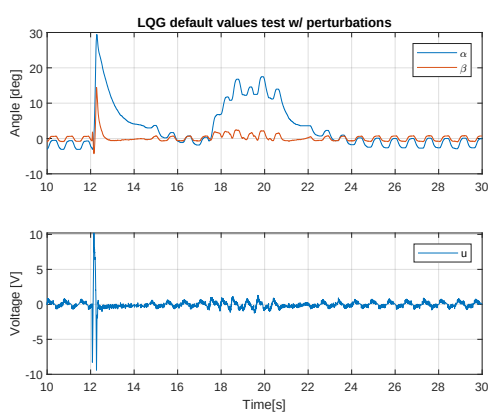


Figura 2: Default LQG response to a physical disturbance.

3) Analysis and Problems: An analysis of this drift revealed two coupled causes rooted in the hardware's physical properties and the controller's structure:

- Motor Deadzone: The DC motor only actuates when the applied voltage exceeds a specific threshold. Within this deadzone range, the control inputs (u) calculated by the LQR do absolutely nothing. As a consequence, the rotary arm will always exhibit a large amplitude of oscillation. The system drifts until the position error grows large enough for the controller to finally command a voltage that breaks through the deadzone, causing the motor to suddenly move, overshoot, and repeat the cycle.
- Absence of an Integrative State: The problem of the steady-state drift is fundamentally tied to the composition of our state vector. The standard LQR model relies strictly on the proportional feedback of the position and velocity states. Because it lacks an integrative state dedicated to accumulating the error of α over time, the controller has no "memory."It inherently accepts steady-state offsets whenever the proportional control effort generated by that offset is smaller than the deadzone resistance, causing the arm to settle away from the reference point.

B. Propused Solutions and Simulation

C. Final Hardware Testing and Validation

IV. Conclusion

V. Anexos